

Subtree Queries

Time limit: 1.00 s

Memory limit: 512 MB

You are given a rooted tree consisting of n nodes. The nodes are numbered $1, 2, \dots, n$, and node 1 is the root. Each node has a value.

Your task is to process following types of queries:

1. change the value of node s to x
2. calculate the sum of values in the subtree of node s

Input

The first input line contains two integers n and q : the number of nodes and queries. The nodes are numbered $1, 2, \dots, n$.

The next line has n integers $v_1, v_2, \dots, v_n$: the value of each node.

Then there are $n - 1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Finally, there are q lines describing the queries. Each query is either of the form "1 s x " or "2 s ".

Output

Print the answer to each query of type 2.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq a, b, s \leq n$
- $1 \leq v_i, x \leq 10^9$

Example

Input	Output
5 3 4 2 5 2 1 1 2 1 3 3 4 3 5 2 3 1 5 3 2 3	8 10